FYUGP B.Sc. PHYSICS HONOURS

ELECTRODYNAMICS II

Semester V · Core in Major

MODULE 2: ELECTROMAGNETIC INDUCTION AND MAXWELL'S EQUATIONS

Past questions

Section A: 1 to 3 marks

Ohm's law and emf

1 Explain Ohm's law. Discuss the terms involved. **(3 marks)** *April 2022*

Faraday's law and induced electric field

2 State and explain Faraday's law. **(2 marks)** *November 2019*

3 State and explain Faraday's law in electromagnetic induction. **(2 marks)** *November 2021*

Maxwell's equations and displacement current

4 Displacement current is given by the equation _____, if the electric field is $\vec{E}$. **(1 mark)** *November 2019*

5 Give the Maxwell's equations in linear medium. **(2 marks)** *November 2019*

6 Magnetic field can be produced by a current or by a changing _____. **(1 mark)** *November 2021*

7 Write down the importance of displacement current in Maxwell's equations. **(2 marks)** *November 2021*

8 What do you mean by the term displacement current ? Give an expression for the same. **(3 marks)** *April 2022*

9 Illustrate the symmetry of Maxwell's equations for $\vec{E}$ and $\vec{B}$ in the absence of the charge and current density terms. **(3 marks)** *April 2022*

Boundary conditions

10 State whether true or false: The tangential component of $\vec{H}$ is continuous across the boundary between two media. **(1 mark)** *November 2021*

11 Write down the boundary conditions for the magnetic field vector $\vec{B}$ at an interface separating two linear media of permittivities ε_1 and ε_2 and permeabilities μ_1 and μ_2. **(3 marks)** *April 2022*

Section B: 4 to 5 marks

Faraday's law and induced electric field

12 An insulated wire has an iron core. An aluminium ring is slipped over the core so as to rest on the top of the coil. If an alternating current is passed through the coil show that the coil will repel the ring. **(4 marks)** *November 2019*

13 A 50 cm. long solenoid having 500 turns and radius 2 cm. is wound on an iron core of relative permeability 800. What will be the average e.m.f. induced in the solenoid if the current in it changes from 0 to 2 A in 0.05 s? **(4 marks)** *April 2020*

Inductance and energy in magnetic fields

14 Find Neumann formula for Mutual induction. **(4 marks)** *November 2019*

15 A long coaxial cable has current I flowing down the surface of the inner cylinder of radius a, and back along the outer cylinder of radius b. Find the magnetic energy stored in section of length l. **(4 marks)** *November 2019*

16 Write down Neumann's formula for mutual induction. Explain its importance. **(4 marks)** *November 2021*

17 Find the magnetic flux through a solenoid of length l with number of turns per unit length N and radius R carrying a current I. Also calculate self inductance per unit length of the coil. **(4 marks)** *November 2021*

18 Find the energy stored in a long coaxial cable of length l which carries a current I flowing down the surface of the inner cylinder of radius a and back along the outer cylinder of radius b. **(4 marks)** *November 2021*

19 Obtain an expression for the energy stored in a magnetic field in terms of current. **(4 marks)** *November 2021*

20 Obtain an expression for the energy stored in a magnetic field due to a current. **(5 marks)** *April 2022*

Maxwell's equations and displacement current

21 Why Ampere Circuital law is to be modified? How is it modified by Maxwell? **(4 marks)** *November 2019*

22 Write down the integral forms of Maxwell's equations and explain the terms involved. **(5 marks)** *April 2022*

Boundary conditions

23 Explain magneto static boundary condition. **(4 marks)** *April 2018*

24 Are the electric field and magnetic field continuous at a boundary between two different media ? Find the relevant equations. **(4 marks)** *November 2019*

25 Write down the boundary conditions for the electric and magnetic vectors $\vec{E}$ and $\vec{B}$ at an interface separating two linear dielectrics of permittivities ε_1 and ε_2 and permeabilities μ_1 and μ_2. **(4 marks)** *November 2021*

Section C: 10 marks

Inductance and energy in magnetic fields

26	Discuss the growth of current in an inductor in a circuit containing resistance connected to a cell of steady e.m.f. Also find the time constant of the circuit. **(10 marks)**	*November 2019*
27	Describe the Raleigh's method to find the self inductance of a coil with the necessary diagram and theory. **(10 marks)**	*November 2019*
28	Discuss the growth of current in a circuit containing an inductance L and a resistance R connected in series with a cell of steady e.m.f. Explain the time constant of the circuit. **(10 marks)**	*November 2021*

Maxwell's equations in matter

29	Explain how Maxwell's equations in vacuum are modified in Matter. **(10 marks)**	*November 2019*
30	Derive the Maxwell's equations inside a polarized matter. **(10 marks)**	*November 2021*

Boundary conditions

31	Using suitable figures, discuss the magnetostatic boundary conditions. **(10 marks)**	*November 2022*